

Non-Repudiation -- Guided Notes ANSWER KEY

Unit tp-6.8 | CompTIA Network+ N10-009 Obj. FC0-U71 6.1 | N10-008 FC0-U71 6.1

For instructor use / distribute after students complete the notes.

Question 1

Non-repudiation is the security property that ensures a party _____ deny having sent a message or performed an action after the fact. It provides _____ of origin (sender cannot deny sending) and _____ of receipt (receiver cannot deny receiving). It is achieved through a combination of _____ and _____.

Answer: Cannot; proof; proof; authentication; audit logging.

Question 2

Digital signatures provide non-repudiation for electronic documents. The sender uses their _____ key to sign the hash of the document. The receiver uses the sender's _____ key to verify the signature. Only the sender's private key could have produced that signature, so the sender _____ deny signing the document.

Answer: Private; public; cannot.

Question 3

A Certificate Authority (CA) is a trusted third party that issues _____ certificates that bind a public key to an identity. This is part of the PKI (Public Key Infrastructure). When a CA signs a certificate, it provides non-repudiation for _____ transactions by confirming that the public key genuinely belongs to the stated _____.

Answer: Digital; electronic; identity (entity).

Question 4

A digital signature is _____ from a written signature because it is cryptographically bound to the _____ of the document. If the document changes after signing, the signature becomes _____. A written signature _____ detect document tampering after the fact.

Answer: Different (stronger); content; invalid; cannot.

Question 5

In legal and financial contexts, non-repudiation ensures that electronic contracts and transactions are _____. The US E-SIGN Act and UETA give electronic signatures the same _____ weight as handwritten signatures. eSignature platforms like DocuSign use PKI-based digital signatures to provide court-admissible _____.

Answer: Enforceable (legally binding); legal; non-repudiation.

Question 6

Timestamping adds non-repudiation for _____ -- proving that a signature or document existed at a specific point in time. A trusted timestamping authority (TSA) signs a hash of the document along with

the current _____, creating a verifiable record that the document existed in its current form at that _____.

Answer: Time; timestamp; moment.

Question 7

HMAC (Hash-based Message Authentication Code) provides _____ and data _____, but not non-repudiation. This is because HMAC uses a _____ key -- both sender and receiver have the same key, so either party could have created the MAC. Non-repudiation requires _____ keys (only the sender's private key can sign).

Answer: Authentication; integrity; shared (symmetric); asymmetric.

Question 8 -- Real World Application

REAL WORLD: A contractor submits an invoice via email claiming they delivered 200 hours of work. The company disputes receiving the deliverables and refuses to pay. The contractor says they sent the files; the company says they never arrived. There is no digital signature on the email. Explain what non-repudiation mechanism would have resolved this dispute and describe how it works.

Answer: A digitally signed email with a trusted timestamp would have resolved this. Here is how it works: (1) The contractor configures their email client with an S/MIME certificate issued by a trusted CA. (2) When they send the deliverables, their email client hashes the email body and attachments, then encrypts the hash with their private key to produce a digital signature. (3) The signed email is sent. The company receives it and their email client verifies the signature using the contractor's public key (obtained from the CA certificate). If verification passes, it proves the email came from the contractor and the content was not modified in transit. (4) A trusted timestamp from a TSA proves the signed message existed at a specific date and time. With this evidence: the contractor cannot deny sending (signature uses their private key), the company cannot deny receiving (the mail server logs and signature receipt are recorded), and the content (deliverables) cannot be disputed (any change would invalidate the signature). This is legally admissible in contract disputes.